

ZALAK KARNIK

Data Scientist | AI Engineer | Generative AI & Machine Learning

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Surat, India · Open to relocation

PROFESSIONAL SUMMARY

Data Scientist and AI Engineer with 2+ years building and deploying production-grade ML and Generative AI systems, spanning RAG pipelines, LLM applications, time-series forecasting, and risk modeling on large-scale data. Proficient in Python, SQL, and cloud platforms (GCP, AWS, Azure), with a track record of taking projects from POC to production and communicating technical trade-offs clearly to stakeholders.

TECHNICAL SKILLS

Languages & Backend: Python, SQL, FastAPI, REST APIs, Streamlit, ETL / Data Pipelines

Machine Learning: scikit-learn, TensorFlow, Keras, Regression, Classification, Clustering, Ensemble Methods, Deep Learning, NLP

Statistics & Optimization: Statistical Modeling, Time-Series Forecasting (Prophet), Hyperparameter Optimization (Optuna), Risk Modeling

Generative AI: RAG Pipelines, LangChain, Agent-based Workflows, Prompt Engineering, Embeddings & Vector Search, Reranking, Text-to-SQL, LLMs (open-source & third-party), llama.cpp

Cloud & MLOps: GCP (Vertex AI, BigQuery), AWS (Bedrock, EC2, S3), Azure (Azure OpenAI), Git, CI/CD, Model Deployment & Monitoring

Data & Search: PostgreSQL, Redis, Elasticsearch (BM25 + Vector Search), BigQuery, Pandas, NumPy

PROFESSIONAL EXPERIENCE

Associate Data Scientist (Gen AI)

Dec 2024 – Present

Casepoint

Surat, India

- Architected and deployed RAG pipelines enabling semantic search, Text-to-SQL querying, and automated document summarization, cutting manual review time by ~50% across 3 production use cases.
- Engineered a hybrid retrieval strategy combining BM25, vector search, and reranking, driving a 40% reduction in irrelevant outputs and improving system precision and reliability.
- Established GenAI architecture standards using open-source and third-party LLMs, LangChain, Redis, and Elasticsearch; served as design authority for model selection and configuration.
- Built a secure, user-configurable prompt-management system (Redis + FastAPI); validated and benchmarked APIs for reliability and communicated architectural trade-offs to cross-functional stakeholders.

Data Science Intern

Jan 2024 – Oct 2024

L&T Finance

Mumbai, India

- Built a time-series forecasting model (Prophet) on GCP Vertex AI, improving loan-disbursement prediction accuracy by 15% and enabling better planning and resource allocation.
- Conducted risk analysis across the loan lifecycle, reducing fraud-detection time by 30% and strengthening response to financial risk signals.
- Managed large-scale financial data in BigQuery (50M+ records); deployed and monitored ML models on Vertex AI across the full model lifecycle.
- Migrated 10+ R scripts to Python, reducing average processing time by ~40% and improving maintainability across the data science team.

KEY PROJECTS

AI Multi-Agent Financial Research Platform · Python, Multi-Agent Orchestration, FastAPI, React · [Live Demo](#)

- Built an end-to-end platform that turns a single stock ticker into a full equity research report for NSE/BSE companies, orchestrating 11 specialized AI agents across fundamentals, valuation, news, competitors, and risk.
- Computed every figure deterministically from real market data with full source traceability, so the system reports data gaps instead of inventing numbers, and added an automated self-review step before publishing.

Chat with Multiple PDFs (Gemini RAG Chatbot) · Python, LangChain, Google Gemini, FAISS, Streamlit

- Built a Streamlit chatbot to upload and query multiple PDFs in natural language, using a retrieval-augmented pipeline with PyPDF2 extraction, chunking, and FAISS vector search for fast, context-aware answers.

Credit Card Fraud Detection · Python, Random Forest, Optuna

- Built a classification pipeline using an ensemble (Random Forest) model with Optuna hyperparameter tuning, improving F1-score from 0.77 to 0.79 and reducing false positives.

Movie Recommendation System · Python, Cosine Similarity, KNN

- Designed a hybrid recommendation engine using content-based filtering (cosine similarity) and collaborative filtering (KNN), improving personalization through retrieval and ranking techniques.

EDUCATION

MSc, Data Science & Big Data Analytics · MIT World Peace University, Pune

2022 – 2024 · CGPA 9.15/10

BSc, Mathematics · Sir P T Sarvajani College of Science, Surat

2019 – 2022 · CGPA 8.50/10

CERTIFICATIONS

AWS Academy Cloud Foundation (2022) · Amazon ML School — Amazon (2024)